

Notes: Massa, Zhang, and Zhang (RFS, 2015)

The Invisible Hand of Short Selling: Does Short Selling Discipline Earnings Management?

Contents

1	Big picture	3
2	Data and variable construction	3
2.1	International sample and data sources	3
2.2	Earnings management: total accruals	4
2.3	Modified Jones residual accruals	4
2.4	Short-selling potential	4
2.5	Control variables	5
3	Models and empirical specifications	5
3.1	Baseline panel regression	5
3.2	Disciplining hypothesis versus price-pressure hypothesis	5
3.3	Instrumental variables using ETF ownership	6
3.4	Testing exclusion restrictions	6
3.5	U.S. Regulation SHO experiment	6
3.6	Hong Kong short-selling eligibility experiment	6
3.7	Global short-selling regulations	7
3.8	Stock price informativeness channel	7
4	Identification and empirical design	7
4.1	Why focus on short-selling potential instead of actual shorting?	7
4.2	Main endogeneity concerns	7
4.3	Instrumental-variable logic	8
4.4	Regulation SHO logic	8
4.5	Hong Kong eligibility-list logic	8
4.6	Limits of the evidence	8
5	Tables	9
5.1	Table 1: Summary statistics and correlations	9
5.2	Table 2: Baseline short-selling potential and earnings management	9
5.3	Table 3: Instrumental-variable approach	10
5.4	Table 4: U.S. Regulation SHO experiment	11
5.5	Table 5: Hong Kong short-selling eligibility list	12
5.6	Table 6: Subsample analyses	12
5.7	Table 7: Robustness checks	12
5.8	Table 8: Stock price informativeness and earnings management	13
6	Short summary	14
7	Conclusion	14
7.1	Core conclusion	14

7.2	Economic intuition	15
7.3	Takeaway	15

1 Big picture

- **Question.** Does the threat of short selling discipline managers and reduce earnings management, or does it instead increase managers' incentives to manipulate earnings to avoid negative price pressure?
- **Core hypothesis.** The disciplining hypothesis says that short sellers increase the probability and speed with which earnings manipulation is revealed. Anticipating this monitoring, managers reduce discretionary accruals.
- **Alternative hypotheses.** The price-pressure hypothesis says managers may manipulate more aggressively to avoid attracting short sellers or suffering bear raids. The ignorance hypothesis says managers behave as if short sellers do not matter.
- **Key explanatory variable.** The paper focuses on *short-selling potential* (SSP), not realized short sales. The main proxy is *Lendable*: the annual fraction of a firm's shares that are available to be lent to short sellers.
- **Key outcome.** Earnings management is mainly measured by modified Jones residual accruals, *AccrualMJones*. Higher absolute/discretionary accruals indicate greater managerial discretion in reported earnings.
- **Data.** The main international panel covers firms from 33 countries over 2002–2009, combining DataExplorers securities lending data, Datastream/WorldScope accounting and returns, and FactSet/LionShares institutional ownership.
- **Main result.** Firms with greater short-selling potential have lower future earnings management. In the baseline regressions, the coefficient on *Lendable* is negative and statistically significant across pooled, fixed-effect, and dynamic-panel specifications.
- **Identification.** The paper uses several approaches: ETF ownership and industry-level ETF ownership as instruments, the U.S. Regulation SHO randomized pilot, and Hong Kong short-sale eligibility changes.
- **External governance result.** Short selling acts as an external governance mechanism: it disciplines managers through the threat that informed traders will detect and trade against manipulation.
- **Additional channel.** Short-selling potential improves stock price informativeness partly because it reduces earnings management, not merely because it mechanically improves trading efficiency.
- **Takeaway.** Short sellers affect not only prices but also real managerial reporting behavior. The possibility of being shorted can discipline firms before short sellers actually attack.

2 Data and variable construction

2.1 International sample and data sources

- The sample period is 2002–2009.
- The initial universe is all publicly listed companies worldwide with accounting and stock-market information from Datastream/WorldScope.
- The sample is matched to short-selling data from DataExplorers and institutional ownership data from FactSet/LionShares.
- DataExplorers provides stock-level information on shares on loan and shares available to be lent to short sellers.
- The data frequency varies by period: monthly in 2002–2003, weekly from July 2004, and daily after 2006.
- After matching and requiring financial controls, the sample contains 17,555 firms in 33 countries.

- The paper reports that the final sample covers more than 90% of global stock-market capitalization.

Interpretation: The global setting is important because short-selling institutions and securities lending depth vary substantially across firms and countries. This variation helps identify whether potential short selling disciplines earnings management.

2.2 Earnings management: total accruals

The paper starts from total accruals, scaled by lagged total assets:

$$Accruals = \frac{(\Delta CA - \Delta Cash) - (\Delta CL - \Delta SD - \Delta TP) - DP}{A_{t-1}}, \quad (1)$$

where

- ΔCA is the change in current assets;
- $\Delta Cash$ is the change in cash and equivalents;
- ΔCL is the change in current liabilities;
- ΔSD is the change in short-term debt included in current liabilities;
- ΔTP is the change in income taxes payable;
- DP is depreciation and amortization expense;
- A_{t-1} is lagged total assets.

Interpretation: Total accruals contain both normal working-capital accruals and discretionary components. The paper therefore moves from total accruals to residual accruals to isolate managerial discretion.

2.3 Modified Jones residual accruals

The main earnings-management measure is modified Jones residual accruals. A standard country-year regression is conceptually:

$$Accruals_{i,t} = \alpha_{c,t} + \beta_1 \frac{1}{A_{i,t-1}} + \beta_2 \frac{\Delta Sales_{i,t} - \Delta Receivables_{i,t}}{A_{i,t-1}} + \beta_3 \frac{PPE_{i,t}}{A_{i,t-1}} + u_{i,t}. \quad (2)$$

The residual is

$$AccrualMJones_{i,t} = \hat{u}_{i,t}. \quad (3)$$

Interpretation: A larger residual accrual indicates that reported accruals deviate more from what is predicted by normal business activity. This is the paper's main proxy for earnings management.

2.4 Short-selling potential

The main short-selling potential variable is:

$$Lendable_{i,t} = \frac{\text{value of shares supplied to securities lending market}_{i,t}}{\text{market capitalization}_{i,t}}. \quad (4)$$

The paper uses the annual average of the monthly, weekly, or daily ratio.

- *Lendable* measures ex ante potential short-sale pressure rather than realized short interest.
- Realized short selling can be endogenous to observed manipulation.
- Lendable supply captures the maximum feasible pressure short sellers can exert.

Interpretation: The disciplining mechanism depends on the manager’s ex ante perception that short sellers can attack. Therefore, lendable shares are closer to the theoretical object than actual short-sale trades.

2.5 Control variables

The standard controls $X_{i,t}$ include:

- log firm size (*Size*);
- log book-to-market ratio (*BM*);
- financial leverage (*Leverage*);
- annual stock return (*Return*);
- stock return volatility (*STD*);
- ADR indicator;
- MSCI country index membership;
- number of analysts following the firm;
- closely held ownership (*CH*);
- institutional ownership (*IO*);
- Amihud illiquidity (*Illiquidity*).

Interpretation: The controls absorb firm size, valuation, risk, visibility, governance, ownership, and trading-liquidity channels that could jointly affect earnings management and short-selling feasibility.

3 Models and empirical specifications

3.1 Baseline panel regression

The main specification is:

$$AccrualMJones_{i,t+1} = \alpha + \beta_1 Lendable_{i,t} + \beta_2 X_{i,t} + \varepsilon_{i,t}. \quad (5)$$

- The dependent variable is measured in year $t + 1$.
- The key explanatory variable and controls are measured in year t .
- Baseline fixed effects include industry, country, and year fixed effects.
- Additional specifications use firm fixed effects and Arellano–Bond dynamic-panel estimation.

Interpretation: The lag structure reduces mechanical simultaneity. A negative β_1 means greater ex ante short-selling potential is associated with less future earnings management.

3.2 Disciplining hypothesis versus price-pressure hypothesis

The competing predictions are:

$$\text{Disciplining hypothesis: } \beta_1 < 0, \quad (6)$$

$$\text{Price-pressure hypothesis: } \beta_1 > 0, \quad (7)$$

$$\text{Ignorance hypothesis: } \beta_1 = 0. \quad (8)$$

Interpretation: The paper’s baseline estimates are strongly consistent with the disciplining hypothesis and inconsistent with the price-pressure and ignorance hypotheses.

3.3 Instrumental variables using ETF ownership

The paper uses ETF ownership and industry-level ETF ownership as instruments for lendable shares. The first stage is:

$$Lendable_{i,t} = \alpha + \pi_1 ETF_{i,t} + \pi_2 X_{i,t} + \eta_{i,t}, \quad (9)$$

or

$$Lendable_{i,t} = \alpha + \pi_1 ETFIndustry_{i,t} + \pi_2 X_{i,t} + \eta_{i,t}. \quad (10)$$

The second stage is:

$$AccrualMJones_{i,t+1} = \alpha + \beta_1 \widehat{Lendable}_{i,t} + \beta_2 X_{i,t} + \varepsilon_{i,t}. \quad (11)$$

Interpretation: ETFs mechanically increase lendable supply because ETF shares are often held by institutions engaged in securities lending. The identifying assumption is that ETF ownership affects earnings management mainly through short-selling potential, not through a direct monitoring or accounting channel.

3.4 Testing exclusion restrictions

The paper checks whether ETF ownership directly predicts accruals in subsamples where short selling is limited:

- countries where short selling is illegal;
- periods with short-sale bans;
- firms with very low lendable shares.

Interpretation: If ETF ownership had a direct governance effect, it should predict accruals even when short selling cannot operate. The exclusion tests largely weaken the concern that ETF ownership directly affects earnings management.

3.5 U.S. Regulation SHO experiment

The U.S. Regulation SHO pilot randomly selected Russell 3000 stocks for removal of the uptick rule. The panel specification is:

$$AccrualMJones_{i,t+1} = \alpha + \beta_1 USSHO_i \times D_{2005-2007,t} + \beta_2 USSHO_i \times D_{2008-2009,t} + \beta_3 X_{i,t} + \varepsilon_{i,t}. \quad (12)$$

A differences specification compares before and after averages:

$$\Delta AccrualMJones_i = \alpha + \beta_1 USSHO_i + \beta_2 \Delta X_i + \varepsilon_i. \quad (13)$$

Interpretation: Because the pilot stocks were selected randomly, the SHO experiment provides a quasi-experimental test of whether relaxing short-sale constraints reduces earnings management.

3.6 Hong Kong short-selling eligibility experiment

Hong Kong regulators changed the list of stocks eligible for short selling. The panel specification is:

$$AccrualMJones_{i,t+1} = \alpha + \beta_1 HKSS_{i,t} + \beta_2 X_{i,t} + \varepsilon_{i,t}, \quad (14)$$

where $HKSS_{i,t} = 1$ if the stock is eligible for short selling.

The change specification is:

$$\Delta AccrualMJones_{i,t+1} = \alpha + \beta_1 \Delta HKSS_{i,t} + \beta_2 \Delta X_{i,t} + \varepsilon_{i,t}. \quad (15)$$

Interpretation: Inclusion in the eligible short-sale list should reduce earnings management if short-selling eligibility disciplines managers. Exclusion should weaken the discipline.

3.7 Global short-selling regulations

The paper also uses cross-country regulatory variation:

- legality of short selling;
- short-sale bans during the sample;
- feasibility of short selling across regulatory environments.

Interpretation: If short selling disciplines managers, the relation between lendable shares and accruals should be stronger when short selling is legally and operationally feasible.

3.8 Stock price informativeness channel

The paper studies whether short selling increases price informativeness through reduced earnings management. The main specification is:

$$PriceInformativeness_{i,t} = \alpha + \beta_1 Lendable_{i,t} + \beta_2 Lendable_{i,t} \times SSPImpact_{i,t} + \beta_3 SSPImpact_{i,t} + \beta_4 X_{i,t} + \varepsilon_{i,t}. \quad (16)$$

- *SSPImpact* equals one for small firms with positive *AccrualMJones*.
- The informativeness measures are downside-minus-upside R^2 and downside-minus-upside nonsynchronicity.

Interpretation: If the disciplining channel matters, the improvement in price informativeness should be strongest where short selling has the greatest effect on earnings management.

4 Identification and empirical design

4.1 Why focus on short-selling potential instead of actual shorting?

- Actual short positions may be a reaction to manipulation that short sellers already observe.
- Managers respond to the threat of being targeted before short sellers trade.
- Lendable supply captures whether short sellers could attack if they uncover manipulation.

Interpretation: The paper’s economic mechanism is preventive: managers reduce manipulation because short sellers could discover and punish it.

4.2 Main endogeneity concerns

The baseline correlation could be biased because:

- better-governed firms may have both lower accruals and greater lendable supply;
- institutional ownership may affect both lending supply and monitoring;
- firms with more transparent information environments may be easier to lend and less likely to manipulate;
- short-selling potential may be correlated with market development and country-level institutions.

Interpretation: The paper addresses these concerns by using fixed effects, controls, instruments, regulatory experiments, and robustness checks.

4.3 Instrumental-variable logic

ETF ownership can shift lendable supply because ETF shares are often available for securities lending. The key identifying assumptions are:

- ETF ownership is correlated with lendable shares;
- ETF ownership does not directly alter managers' earnings-management choices except through short-selling potential.

Interpretation: The first-stage coefficients are strong, and exclusion tests in low-short-selling environments provide evidence that the instrument is not simply proxying for direct monitoring.

4.4 Regulation SHO logic

- Pilot firms were randomly selected from the Russell 3000.
- The pilot removed the uptick-rule restriction for selected stocks.
- If short selling disciplines managers, pilot firms should reduce earnings management after the rule change.

Interpretation: This design is the cleanest causal test in the paper because assignment to the pilot was random rather than chosen by firms or investors.

4.5 Hong Kong eligibility-list logic

- Hong Kong periodically changed the list of stocks eligible for short selling.
- Inclusion increases the feasible threat of short selling.
- Exclusion should reduce the threat.

Interpretation: The Hong Kong test complements the U.S. test because it uses a different regulatory environment and a direct eligibility list rather than an uptick-rule pilot.

4.6 Limits of the evidence

- Lendable shares are not a perfect measure of actual short-selling pressure.
- ETF instruments may still be correlated with unobserved investor attention or governance.
- International accounting data and accrual measures vary across countries.
- Regulatory experiments are valuable but apply to specific markets and institutional settings.

Interpretation: The strength of the paper lies in the consistency of evidence across many identification strategies, not in any single perfect experiment.

5 Tables

5.1 Table 1: Summary statistics and correlations

Read:

- Panel A shows that the mean modified Jones accrual is close to zero, while the standard deviation is about 0.081.
- The mean lendable share ratio is about 0.065, with large cross-sectional dispersion.
- The sample contains more than 61,000 firm-year observations for the main variables.
- Panel B shows that *Lendable* is negatively correlated with *AccrualMJones*.
- Lendable shares are strongly related to firm size, institutional ownership, MSCI membership, and analyst following.

Economic message: The raw data already point toward the disciplining hypothesis, but the correlations also show why rich controls and identification strategies are necessary.

Table 1
Summary statistics

Panel A: Summary statistics

Variable	N	Mean	STD	10%	25%	Median	75%	90%
<i>AccrualMJones</i>	61,624	0.002	0.081	-0.081	-0.034	0.003	0.037	0.080
<i>Lendable</i>	61,624	0.065	0.094	0.000	0.004	0.023	0.086	0.211
<i>Size</i>	61,624	12.939	1.821	10.725	11.673	12.803	14.108	15.369
<i>BM</i>	61,624	-0.622	0.879	-1.676	-1.133	-0.593	-0.067	0.419
<i>Leverage</i>	61,624	0.198	0.180	0.000	0.024	0.170	0.320	0.450
<i>Return</i>	61,624	0.023	0.649	-0.752	-0.241	0.094	0.373	0.678
<i>STD</i>	61,624	0.448	0.316	0.189	0.258	0.372	0.548	0.776
<i>ADR</i>	61,624	0.036	0.185	0.000	0.000	0.000	0.000	0.000
<i>MSCI</i>	61,624	0.663	0.473	0.000	0.000	1.000	1.000	1.000
<i>Analyst</i>	61,624	5.053	5.998	0.000	1.000	2.833	7.500	13.417
<i>CH</i>	61,624	0.311	0.241	0.002	0.109	0.285	0.491	0.653
<i>IO</i>	61,624	0.248	0.299	0.000	0.020	0.110	0.379	0.785
<i>Illiquidity</i>	61,624	-3.430	2.965	-7.353	-5.641	-3.371	-1.284	0.407

(continued)

Table 1
Continued

Panel B: Correlation Coefficients (Spearman for the upper-right part, highlighted; Pearson for the bottom-left part)

Variable	<i>AccrualMJones</i>	<i>Lendable</i>	<i>Size</i>	<i>BM</i>	<i>Leverage</i>	<i>Return</i>	<i>STD</i>	<i>ADR</i>	<i>MSCI</i>	<i>Analyst</i>	<i>CH</i>	<i>IO</i>	<i>Illiquidity</i>
<i>AccrualMJones</i>	-0.030	-0.001	0.004	0.002	0.015	-0.041	-0.042	-0.003	-0.011	-0.014	-0.029	0.011	-0.001
<i>Lendable</i>	0.501	-0.170	0.013	-0.026	-0.135	0.074	0.344	0.377	-0.219	0.352	-0.219	0.352	-0.338
<i>Size</i>	0.020	0.319	0.378	0.197	0.143	-0.341	0.361	0.798	0.732	-0.066	0.383	-0.066	0.383
<i>BM</i>	0.015	-0.114	-0.204	0.085	-0.204	-0.080	-0.042	-0.103	-0.245	0.097	-0.215	0.258	-0.015
<i>Leverage</i>	0.020	0.008	0.122	0.016	0.006	-0.099	0.022	-0.114	0.095	0.067	-0.017	-0.154	0.020
<i>Return</i>	0.023	-0.028	0.146	-0.199	-0.023	-0.002	0.003	0.077	-0.007	0.035	0.014	-0.034	-0.023
<i>STD</i>	-0.040	-0.110	-0.300	-0.097	-0.093	0.068	-0.031	-0.148	-0.178	-0.039	-0.072	0.243	-0.040
<i>ADR</i>	-0.046	0.051	0.101	-0.036	0.017	0.001	-0.019	0.002	0.143	-0.063	0.004	-0.116	-0.046
<i>MSCI</i>	-0.004	0.246	0.750	-0.090	0.006	0.002	-0.160	0.003	0.600	-0.027	0.208	-0.025	-0.004
<i>Analyst</i>	-0.011	0.348	0.726	-0.212	0.066	-0.013	-0.174	0.109	0.144	-0.112	0.043	-0.091	-0.011
<i>CH</i>	-0.016	-0.255	-0.084	0.073	0.007	0.039	-0.038	-0.005	-0.049	-0.134	-0.091	-0.091	-0.016
<i>IO</i>	0.007	0.403	0.952	-0.207	-0.065	-0.007	-0.070	-0.048	0.261	0.358	-0.219	-0.409	-0.007
<i>Illiquidity</i>	-0.022	-0.384	-0.068	0.235	-0.132	-0.038	0.249	-0.123	-0.609	-0.075	0.229	-0.409	-0.022

Table 1, Panel A: summary statistics

Table 1, Panel B: correlations

5.2 Table 2: Baseline short-selling potential and earnings management

Read:

- The coefficient on *Lendable* is negative in all specifications.
- In the baseline specification, the coefficient is about -0.044 with a t -statistic of -7.88 .
- Firm fixed-effects specifications and Arellano–Bond specifications also deliver negative coefficients.
- The result survives controls for size, book-to-market, leverage, returns, volatility, ADR, MSCI membership, analysts, ownership, and illiquidity.

Economic message: Greater short-selling potential predicts lower future discretionary accruals. This is the core baseline evidence that short sellers discipline managers.

Table 2
Short selling and earnings management

	Baseline Model		Firm Fixed Effects		Arellano-Bond	
	Model (1)	Model (2)	Model (3)	Model (4)	Model (5)	Model (6)
<i>Lendable</i>	-0.044 (-7.88)	-0.032 (-7.10)	-0.017 (-2.33)	-0.052 (-8.39)	-0.041 (-7.38)	-0.032 (-7.01)
<i>Lagged Accrual</i> <i>MJones</i>					-0.010 (-1.02)	-0.009 (-1.00)
<i>Size</i>	0.004 (7.10)	0.005 (8.09)	0.022 (12.44)	0.021 (12.14)	0.003 (5.98)	0.004 (6.99)
<i>BM</i>	0.004 (6.96)	0.005 (7.60)	0.013 (8.15)	0.013 (8.63)	0.004 (7.02)	0.005 (7.74)
<i>Leverage</i>	0.010 (4.51)	0.010 (4.52)	0.070 (10.04)	0.071 (10.22)	0.011 (4.65)	0.011 (4.71)
<i>Return</i>	0.003 (3.63)	0.001 (1.99)	-0.000 (-0.26)	-0.000 (-0.49)	0.003 (3.90)	0.002 (2.42)
<i>STD</i>	-0.006 (-3.85)	-0.004 (-2.81)	-0.009 (-3.54)	-0.007 (-2.90)	-0.008 (-4.55)	-0.006 (-3.59)
<i>ADR</i>	-0.013 (-5.82)	-0.014 (-6.08)	0.002 (0.32)	0.005 (0.69)	-0.013 (-5.96)	-0.014 (-6.19)
<i>MSCI</i>	-0.006 (-5.57)	-0.006 (-5.85)			-0.006 (-5.62)	-0.006 (-5.70)
<i>Analyst</i>	-0.001 (-10.29)	-0.001 (-10.51)	-0.001 (-3.63)	-0.000 (-1.52)	-0.001 (-10.33)	-0.001 (-10.35)
<i>CH</i>	-0.004 (-2.19)	-0.004 (-2.26)	0.002 (0.42)	0.000 (0.09)	-0.004 (-2.10)	-0.004 (-2.21)
<i>IO</i>	0.003 (1.26)	0.003 (1.35)	-0.005 (-0.83)	-0.006 (-0.87)	0.003 (1.54)	0.004 (1.67)
<i>Illiquidity</i>	-0.001 (-2.48)	-0.000 (-1.35)	-0.003 (-4.16)	-0.001 (-2.12)	-0.001 (-3.76)	-0.001 (-2.61)
Fixed Effects	ICY	IC	FY	F	ICY	IC
Obs	61,624	61,624	61,624	61,624	59,446	59,446
AdjRsqr	2.9%	2.9%	19.4%	19.1%		

5.3 Table 3: Instrumental-variable approach

Read:

- Panel A uses ETF ownership and industry-level ETF ownership as instruments for lendable shares.
- First-stage coefficients are positive and strong: ETF ownership raises lendable supply.
- The second-stage coefficient on predicted lendable shares is negative.
- With ETF ownership as the instrument, the second-stage coefficient is about -0.102 .
- With industry ETF ownership as the instrument, the second-stage coefficient is about -0.041 .
- Panel B tests exclusion restrictions by checking whether ETF ownership predicts accruals when short selling is unavailable or nearly unavailable.

Economic message: The IV tests support a causal interpretation: shifts in lendable supply induced by ETF ownership are associated with lower earnings management.

Table 3
Instrumental variable approach

A. ETF and industry-level ETF as instrumental variables

Dep. Variable=	Instrument=ETF		Instrument=ETF Industry	
	Lendable (1st Stage)	AccrualMJones (2nd Stage)	Lendable (1st Stage)	AccrualMJones (2nd Stage)
	Model (1)	Model (2)	Model (3)	Model (4)
Instrument	0.847 (5.46)		3.039 (5.30)	
Lendable		-0.102 (-4.69)		-0.041 (-1.86)
Size	-0.004 (-9.08)	0.004 (6.51)	-0.004 (-6.85)	0.004 (6.97)
BM	0.009 (20.86)	0.005 (7.43)	0.008 (19.33)	0.004 (6.61)
Leverage	0.007 (3.90)	0.011 (4.66)	0.004 (2.10)	0.010 (4.50)
Return	0.004 (6.89)	0.003 (3.86)	0.002 (2.99)	0.003 (3.60)
STD	-0.006 (-5.88)	-0.006 (-4.02)	-0.005 (-4.92)	-0.006 (-3.83)
ADR	0.004 (2.48)	-0.013 (-5.67)	-0.002 (-1.23)	-0.013 (-5.84)
MSCI	0.015 (15.71)	-0.005 (-4.48)	0.017 (18.21)	-0.006 (-5.37)
Analyst	0.001 (11.55)	-0.001 (-9.56)	0.001 (9.70)	-0.001 (-10.05)
CH	-0.008 (-5.80)	-0.004 (-2.44)	-0.013 (-8.16)	-0.004 (-2.17)
IO	0.115 (20.03)	0.011 (2.93)	0.132 (40.23)	0.002 (0.61)
Illiquidity	-0.007 (-20.37)	-0.001 (-3.36)	-0.007 (-26.14)	-0.001 (-2.17)
Fixed Effects	ICY	ICY	ICY	ICY
Obs	61,624	61,624	61,624	61,624
AdjRsqr	65.6%	2.8%	67.4%	2.9%

(continued)

Table 3, Panel A: IV estimates

1716

Table 3
B. Tests on Exclusion Restrictions (Regress Accruals on ETFs)

	Accruals on ETFs when SSP is Low						Accruals on Industry ETF when SSP is Low					
	Legality=0		SSShare=1		0 < Lendable < 0.5%		Legality=0		SSShare=1		0 < Lendable < 0.5%	
	Model (1)	Model (2)	Model (3)	Model (4)	Model (5)	Model (6)	Model (7)	Model (8)	Model (9)	Model (10)	Model (11)	Model (12)
ETF	-0.008 (-3.87)	-0.016 (-4.75)	-0.271 (-6.86)	-0.300 (-8.35)	0.010 (0.21)		-0.114 (-2.01)	0.011 (0.16)	-0.037 (-0.23)	0.001 (0.01)	0.266 (1.29)	
ETF Industry												
Lendable												
Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs	61,624	61,624	61,624	61,624	61,624	61,624	61,624	61,624	61,624	61,624	61,624	61,624
AdjRsqr	2.9%	2.9%	8.7%	8.0%	2.7%	2.8%	2.9%	2.9%	6.7%	4.9%	2.8%	

Panel A addresses the endogeneity problem by using ETF ownership (ETF) in industry-level ETF ownership (ETF Industry) as an instrumental variable and presents a panel regression of a firm's earnings management measure (accruals_{it}) on predicted lendable shares (Lendable_{it}) and firm-level control variables (X_{it}) as well as unreported industry, country, and year fixed effects (ICF) on the variation of the following models:

The 1st stage: $Lendable_{it} = \alpha_0 + \beta_1 ETF_{it} + \beta_2 ETF_{Industry} + \beta_3 X_{it} + \epsilon_{it}$

The 2nd stage: $Accruals_{it} = \alpha_0 + \beta_1 Lendable_{it} + \beta_2 Lendable_{Industry} + \beta_3 X_{it} + \epsilon_{it}$

where Lendable_{it} refers to lendable shares and X_{it} includes the list of standard control variables. Models (1) and (3) regress lendable shares on ETF ownership and industry-level ETF ownership, respectively. Models (2) and (4) regress residual accruals on predicted lendable shares. Models (5) and (6) regress residual accruals on the two instruments. Models (7) and (8) also include lendable shares in the same regression. The remaining models regress accruals_{it} on the two instruments for subsamples of the stocks for which short selling is either prohibited (legality=0) or unrestricted (legality=1) or low- (0 < Lendable < 0.5%) or high- (Lendable > 0.5%) liquidity stocks. The statistics reported in parentheses are based on standard errors adjusted for heteroscedasticity and firm-level clustering. N denotes the number of firm-year observations, and AdjRsqr is adjusted R². The sample period is from 2002 to 2009.

Table 3, Panel B: exclusion tests

5.4 Table 4: U.S. Regulation SHO experiment

Read:

- Regulation SHO randomly selected pilot stocks for removal of the uptick-rule restriction.
- Pilot stocks show lower earnings management after the treatment period.
- The coefficient on $USSHO \times Dummy(2005 - 2007)$ is negative and statistically significant.
- The after-versus-before change specifications also show negative effects for pilot stocks.
- The placebo period after the experiment does not show the same strong result.

Economic message: Random assignment in Regulation SHO provides especially persuasive evidence that relaxing short-sale constraints reduces earnings management.

Table 4
The U.S. Regulation SHO Experiment and Earnings Management

Dep. Variable=	AccrualMJones			After=Before=	Δ AccrualMJones (After-Before)	
	2001-2007 Ex. 2004	2001-2009 Ex. 2004	2001-2009 Ex. 2004-2007		2005-2007 2001-2003	2008-2009 2001-2003
	Model (1)	Model (2)	Model (3)		Model (4)	Model (5)
US SHO × Dummy (2005-2007)	-0.006 (-2.20)	-0.006 (-2.16)			-0.005 (-2.05)	-0.003 (-0.88)
US SHO × Dummy (2008-2009)		-0.004 (-0.99)	-0.003 (-0.79)			
Size	0.013 (3.27)	0.011 (3.72)	0.006 (1.69)	Δ Size	0.000 (0.06)	0.001 (0.20)
BM	0.008 (2.54)	0.006 (2.26)	0.001 (0.44)	Δ BM	0.001 (0.32)	-0.004 (-1.31)
Leverage	0.055 (4.84)	0.043 (4.58)	0.027 (2.29)	Δ Leverage	0.008 (0.67)	0.001 (0.10)
Return	0.006 (2.12)	0.004 (1.70)	0.001 (0.40)	Δ Return	0.018 (3.79)	0.003 (0.56)
STD	-0.018 (-3.52)	-0.018 (-4.57)	-0.020 (-4.18)	Δ STD	-0.026 (-3.67)	-0.026 (-3.65)
Analyst	-0.002 (-5.16)	-0.002 (-5.87)	-0.002 (-4.71)	Δ Analyst	-0.001 (-1.81)	-0.001 (-2.17)
CH	-0.001 (-0.08)	-0.003 (-0.45)	-0.007 (-0.86)	Δ CH	0.009 (0.86)	0.000 (0.03)
IO	-0.017 (-1.86)	-0.017 (-2.27)	-0.010 (-0.89)	Δ IO	-0.013 (-1.27)	-0.013 (-1.13)
Illiquidity	-0.006 (-3.63)	-0.006 (-4.16)	-0.005 (-2.92)	Δ Illiquidity	-0.007 (-3.68)	-0.005 (-2.21)
Fixed Effects	FY	FY	FY	Fixed Effects	I	I
Obs	12,597	16,347	10,020	Obs	2,218	2,105
AdjRsqr	15.4%	13.3%	13.8%	AdjRsqr	8.1%	6.6%

This table examines Regulation SHO in the United States, in which the SEC randomly selected a sample of

Table 5
Hong Kong short-selling list and earnings management

Dep. Variable=	AccrualMJones				Δ AccrualMJones	
	Model (1)	Model (2)	HK SS=0 Model (3)		Model (4)	Model (5)
HK SS	-0.022 (-2.18)	-0.024 (-2.25)			-0.023 (-1.98)	
HK SS Post		-0.009 (-0.77)	0.002 (0.12)	ΔHK SS		0.036 (2.25)
Size	0.045 (6.05)	0.045 (6.05)	0.049 (5.48)	Δ Size	0.086 (8.41)	0.086 (8.43)
BM	0.013 (2.00)	0.014 (2.01)	0.015 (1.84)	Δ BM	0.048 (4.17)	0.047 (4.16)
Leverage	0.136 (3.65)	0.136 (3.66)	0.146 (3.18)	Δ Leverage	0.329 (5.62)	0.328 (5.62)
Return	-0.001 (-0.18)	-0.001 (-0.20)	-0.003 (-0.41)	Δ Return	-0.001 (-0.01)	-0.000 (-0.04)
STD	0.003 (0.37)	0.003 (0.37)	0.007 (0.66)	Δ STD	0.011 (0.92)	0.011 (0.93)
ADR	-0.148 (-5.10)	-0.148 (-5.14)	-0.151 (-3.23)	Δ ADR	-0.127 (-1.28)	-0.125 (-1.25)
Analyst	-0.000 (-0.06)	-0.000 (-0.05)	-0.001 (-1.03)	Δ Analyst	-0.001 (-0.89)	-0.001 (-1.07)
CH	0.018 (0.78)	0.018 (0.78)	0.033 (1.14)	Δ CH	0.020 (0.65)	0.020 (0.65)
Illiquidity	0.002 (0.92)	0.002 (0.91)	0.001 (0.42)	Δ Illiquidity	0.001 (0.37)	0.001 (0.34)
Fixed Effects	FY	FY	FY	Fixed Effects	CY	CY
Obs	4,411	4,411	3,483	Obs	3,528	3,528
AdjRsqr	10.8%	10.7%	10.3%	AdjRsqr	6.2%	6.2%

This table explores the unique regulatory setting in the Hong Kong market in which regulators changed the list

5.5 Table 5: Hong Kong short-selling eligibility list

Read:

- Stocks eligible for short selling in Hong Kong have lower modified Jones accruals.
- The coefficient on $HKSS$ is negative in the panel regressions.
- Net inclusion in the short-selling list is associated with lower accruals.
- Exclusion from the list is associated with a positive change in accruals.

Economic message: The Hong Kong setting shows that direct eligibility for short selling disciplines earnings management outside the U.S. institutional environment.

5.6 Table 6: Subsample analyses

Read:

- Panel A shows that the negative relation between lendable shares and accruals holds in different periods and across U.S. and non-U.S. samples.
- The effect appears stronger in the U.S. subsample but remains negative outside the U.S.
- Panel B examines firms with positive versus negative accruals and small versus large firms.
- The disciplining effect is strongest for small firms with positive modified Jones accruals.

Economic message: Short selling disciplines earnings management most where manipulation incentives and informational opacity are likely to be largest.

Table 6 Subsample analyses				
A. Subsample analyses on different time periods and regions				
	≥ 2005	Full Sample	U.S.	N.U.S.
	Model (1)	Model (2)	Model (3)	Model (4)
<i>Lendable</i>	-0.045	-0.003	-0.064	-0.044
	(-6.37)	(-0.14)	(-7.39)	(-3.52)
<i>Lendable $\times T$</i>		-0.006		
		(-2.17)		
<i>Firm Controls and Constant</i>	Yes	Yes	Yes	Yes
Fixed Effects	ICY	ICY	ICY	ICY
Obs	44,171	61,624	21,825	39,799
AdjRsqr	2.7%	2.9%	3.3%	2.8%
B. Subsample analyses on different degrees of earnings management				
	<i>Accrual_{M Jones} ≥ 0</i>		<i>Accrual_{M Jones} < 0</i>	
	Small Firms Model (1)	Large Firms Model (2)	Small Firms Model (3)	Large Firms Model (4)
<i>Lendable</i>	-0.097	-0.025	-0.038	0.004
	(-4.27)	(-3.66)	(-1.55)	(0.55)
<i>Firm Controls and Constant</i>	Yes	Yes	Yes	Yes
Fixed Effects	ICY	ICY	ICY	ICY
Obs	6,022	26,446	6,224	22,932
AdjRsqr	6.9%	8.7%	13.3%	10.0%

5.7 Table 7: Robustness checks

Read:

- Panel A uses alternative short-selling potential measures, including shares on loan, lending fees, fee volatility, specialness, and lending constraints.
- Panel B uses target-beating and earnings-persistence measures of earnings management.
- Panel C uses alternative accrual measures and misstatement/scandal probabilities.
- Panel D controls for alternative governance and discipline channels.

- Across these checks, the disciplining role of short selling remains robust.

Economic message: The results are not driven by one accrual measure or one proxy for short-sale feasibility. The evidence survives a broad set of alternative definitions and controls.

Table 7
Robustness checks

A. Alternative SSP measures

SSP=	<i>On Loan</i>	<i>Fee</i>	<i>STD Fee</i>	<i>Specialness</i>	<i>Constraint</i>
	Model (1)	Model (2)	Model (3)	Model (4)	Model (5)
<i>SSP</i>	-0.052 (-4.12)	0.001 (3.74)	0.003 (4.93)	0.004 (4.05)	0.004 (2.35)
<i>Firm Controls and Constant</i>	Yes	Yes	Yes	Yes	Yes
Fixed Effects	ICY	ICY	ICY	ICY	ICY
Obs	61,623	61,575	59,011	61,575	61,574
AdjRsqr	2.9%	2.9%	2.9%	2.9%	2.9%

B. Alternative earnings management measures on target beating and earnings persistence

Dep. Variable=	Target beating			Earnings persistence	
	<i>SPAF</i>	<i>SPDE</i>	<i>SPE</i>	<i>Earnings</i>	<i>Accrual_{MJones}</i>
	Model (1)	Model (2)	Model (3)	Model (4)	Model (5)
<i>Lendable</i>	-0.767 (-3.63)	-1.111 (-3.99)	-0.914 (-2.06)	0.033 (4.50)	-0.019 (-3.73)
<i>Earnings</i>				0.805 (12.68)	
<i>Lendable × Earnings</i>				-0.301 (-4.66)	
<i>Accrual_{MJones}</i>					0.109 (0.96)
<i>Lendable × Accrual_{MJones}</i>					-0.453 (-4.15)
<i>Firm Controls and Constant</i>	Yes	Yes	Yes	Yes	Yes
<i>Earnings (Accrual_{MJones}) × Control</i>	No	No	No	Yes	Yes
Fixed Effects	ICY	ICY	ICY	ICY	ICY
Obs	46,381	35,986	19,091	58,302	55,816
AdjRsqr (PseRsqr)	3.4%	4.6%	10.2%	68.1%	5.5%

(continued)

Table 7, Panels A–B: alternative SSP and earnings-management measures

Table 7
(continued)

C. Alternative earnings management measures on other accruals and earnings misstatements

Dep. Variable=	Other accrual measures			Misstatement and scandals	
	<i>Accrual_{MJones}</i>	<i>Accrual_{KDJW}</i>	<i>Accrual_{DD}</i>	<i>Prob(Misstatement)</i>	<i>Prob(Scandals)</i>
	Model (1)	Model (2)	Model (3)	Model (4)	Model (5)
<i>Lendable</i>	-0.038 (-6.95)	-0.035 (-4.15)	-0.019 (-3.83)	-0.818 (-2.23)	-1.209 (-2.58)
<i>Firm Controls and Constant</i>	Yes	Yes	Yes	Yes	Yes
Fixed Effects	ICY	ICY	ICY	ICY	ICY
Obs	61,562	61,015	57,603	30,047	30,047
AdjRsqr (PseRsqr)	3.1%	0.5%	1.2%	5.8%	17.6%

D. Alternative discipline channels

	Model (1)	Model (2)	Model (3)	Model (4)	Model (5)
<i>Lendable</i>	-0.027 (-3.06)	-0.028 (-3.04)	-0.032 (-3.43)	-0.033 (-3.62)	-0.022 (-2.32)
<i>ISS</i>	-0.025 (-2.95)	-0.025 (-2.97)	-0.025 (-2.97)	-0.025 (-2.94)	-0.023 (-2.52)
<i>IAS</i>		-0.000 (-0.00)	-0.001 (-0.15)	-0.001 (-0.23)	0.001 (0.29)
<i>BigN</i>			-0.008 (-3.97)	-0.008 (-4.13)	-0.005 (-2.22)
<i>NewsCoverage</i>				-0.002 (-3.16)	-0.003 (-3.75)
<i>Disp</i>					-0.308 (-3.31)
<i>Firm Controls and Constant</i>	Yes	Yes	Yes	Yes	Yes
Fixed Effects	ICY	ICY	ICY	ICY	ICY
Obs	16,184	16,149	16,065	16,065	13,396
AdjRsqr	4.7%	4.7%	4.8%	4.8%	6.1%

Table 7, Panels C–D: alternative accruals and channels

5.8 Table 8: Stock price informativeness and earnings management

Read:

- The paper measures price informativeness using downside-minus-upside R^2 and downside-minus-upside nonsynchronicity.
- Lendable shares are associated with more informative prices.
- The interaction between *Lendable* and *SSPImpact* strengthens the informativeness effect.
- *SSPImpact* is one for small firms with positive modified Jones accruals, where short selling has the greatest disciplining impact.

Economic message: Short selling improves price informativeness partly by reducing firms' incentives to manage earnings. The channel is not only trading efficiency but also better corporate reporting behavior.

Table 8
Stock price informativeness and earnings management

Price Informativeness=	R^2_{DMU}		$NonsynDMU$	
	Model (1)	Model (2)	Model (3)	Model (4)
<i>Lendable</i>	-0.055 (-13.67)	-0.053 (-13.08)	0.383 (9.37)	0.371 (9.42)
<i>Lendable</i> $\times$ <i>SSP</i> _{Impact}		-0.078 (-5.23)		0.455 (1.76)
<i>SSP</i> _{Impact}		0.002 (1.56)		-0.007 (-0.47)
<i>Size</i>	-0.002 (-2.93)	-0.001 (-2.87)	0.014 (2.79)	0.014 (2.83)
<i>BM</i>	-0.001 (-1.49)	-0.001 (-1.33)	0.006 (1.45)	0.006 (1.34)
<i>Leverage</i>	0.003 (1.62)	0.003 (1.59)	-0.024 (-1.37)	-0.024 (-1.36)
<i>Return</i>	0.005 (7.58)	0.005 (7.41)	-0.055 (-8.02)	-0.054 (-7.91)
<i>STD</i>	-0.005 (-4.97)	-0.005 (-4.95)	-0.009 (-0.73)	-0.009 (-0.73)
<i>ADR</i>	-0.003 (-1.55)	-0.003 (-1.52)	0.029 (1.77)	0.028 (1.74)
<i>MSCI</i>	-0.001 (-0.91)	-0.001 (-0.84)	0.036 (3.76)	0.036 (3.72)
<i>Analyst</i>	0.000 (0.05)	0.000 (0.00)	-0.001 (-1.24)	-0.001 (-1.24)
<i>CH</i>	-0.007 (-4.28)	-0.006 (-4.16)	0.074 (4.81)	0.074 (4.75)
<i>IO</i>	0.013 (11.36)	0.013 (11.23)	-0.126 (-11.09)	-0.125 (-11.04)
<i>Illiquidity</i>	0.001 (3.08)	0.001 (3.13)	-0.012 (-3.73)	-0.013 (-3.76)
Fixed Effects	ICY	ICY	ICY	ICY
Obs	59,953	59,953	59,952	59,952
AdjRsqr	7.6%	7.6%	5.4%	5.4%

6 Short summary

- The paper asks whether short selling disciplines earnings management.
- The main measure of short-selling potential is the fraction of shares available to lend, *Lendable*.
- The main earnings-management measure is modified Jones residual accruals.
- A large international panel from 33 countries shows a strong negative relation between *Lendable* and future discretionary accruals.
- ETF ownership and industry-level ETF ownership are used as instruments for lendable shares.
- Regulation SHO and Hong Kong short-sale eligibility changes provide event-based identification.
- The results are strongest for small firms and firms with positive accruals.
- Robustness checks use alternative short-sale measures, alternative earnings-management measures, and controls for other governance channels.
- Short-selling potential also improves stock price informativeness, especially where its disciplining effect on earnings management is strongest.
- The overall conclusion is that short selling is an external governance mechanism, not merely a price-discovery mechanism.

7 Conclusion

7.1 Core conclusion

Massa, Zhang, and Zhang show that the threat of short selling reduces earnings management. Firms whose shares are more available for lending report lower future discretionary accruals. This result survives controls, fixed effects, instrumental variables, U.S. randomized regulatory variation, Hong Kong short-sale eligibility variation, and many robustness checks.

7.2 Economic intuition

Managers manipulate earnings when the private benefit of manipulation exceeds the expected cost of detection. Short sellers change this trade-off because they specialize in finding and trading against bad news. Greater short-selling potential raises the probability and speed with which manipulation is uncovered, so managers reduce manipulation before short sellers actually intervene.

7.3 Takeaway

The paper expands the role of short sellers from information intermediaries in secondary markets to external corporate monitors. Short selling improves market efficiency not only because traders incorporate information into prices, but also because the possibility of short selling changes managers' reporting behavior at the source.